Analytical Model for No-Load Electromagnetic Performance Prediction of V-Shape IPM Motors Considering Nonlinearity of Magnetic Bridges

Shiqi Li, Wenming Tong, *Member, IEEE*, Mingjun Hou, Shengnan Wu, *Member, IEEE*, and Renyuan Tang

***Abstract*—This paper proposes a novel analytical model which combines subdomain method and magnetic equivalent circuit (MEC) method to predict the no-load performance of V-shape interior permanent magnet (IPM) motors with any slot-pole combination considering the nonlinearity of magnetic bridges (MBs). The V-shape magnets are equivalent to the form of distribution along radial and tangential directions, and the motor is divided into two corresponding structures to facilitate the establishment of subdomains. To consider the saturation of MBs, the MEC method are employed to obtain the permeability of MBs to establish the boundary conditions. The no-load magnetic field of slotless motor can be obtained by solving the equations established by boundary conditions, and the conformal mapping is used to consider the effect of slotting. The proposed model can be used to calculate the no-load magnetic field, back electromotive force, and cogging torque of IPM motor. The results are verified by finite element analysis (FEA) and experiments, which confirms the validity of the model for facilitating the motor design and optimization. Compared with FEA, the proposed method has the advantages of faster modeling, shorter time consuming and meanwhile achieves the approximate accuracy. In addition, U-shape motors with MBs can also be analyzed by this model.**

***Index Terms*—Interior permanent magnet motors, subdomain method, magnetic equivalent circuit, magnetic bridge.**

I. Introduction

INTERIOR permanent magnet (IPM) motors are used in a wide range of industrial applications and smart home due to their high-power density, wide speed range, and high reluctance torque compared with surface-mounted permanent magnet (SPM) motors [1]–[3]. Accurate magnetic field calculation of motors is the premise to predict the electromagnetic performance and judge the design rationality [4]. However, due to the complex and changeable rotor configurations of the IPM motors, it is difficult to directly calculate the magnetic field distribution by a general analytical model, which is not conducive to the design and optimization of the IPM motors. Finite element analysis (FEA) is powerful enough to accurately predict the magnetic field distribution of IPM motors. However, it is time-consuming in the initial design stage due to the diversity of rotor configurations and need for high-precision mesh generation [5]–[6]. The analytical method is an efficient tool for obtaining the magnetic field distribution quickly and accurately. Several analytical methods for the prediction of magnetic field distribution in permanent magnet motors have been derived, such as subdomain method based on the Laplace-Poisson equations [7]–[13], magnetic equivalent circuit (MEC) method based on the flux tube theory [14]–[20], and analytical method based on the winding function [21]–[22].

The subdomain method is a method to solve the field governing equations of each subdomain by the boundary conditions between different subdomains. In 1993, Z. Q. Zhu proposed this method to calculate the magnetic field distribution of SPM motors in [7], [8]. Based on the subdomain method, the magnetic field distribution of axial flux permanent magnet (AFPM) motors is calculated in [9]. Considering the changes in boundary conditions caused by changes of rotor configuration in IPM motors, references [10]–[13] calculate the open-circuit magnetic field distribution of surface inset permanent magnet motor, V-shape IPM motor, U-shape IPM motor, and spoke-type permanent magnet (STPM) motor.

However, the subdomain method mentioned above ignore the influence of core saturation, which will cause errors in the calculated magnetic field distribution, especially for the IPM motors with severely saturated magnetic bridges (MBs). The MEC method is adopted by some researchers to consider the influence of core saturation. The three-dimensional MEC is used to obtain the magnetic field distribution of a linear motor in [14]; reference [15] uses the quasi-three-dimensional MEC to calculate the magnetic field distribution of AFPM motors; and the magnetic field distribution of IPM motors is calculated by MEC method in [16]–[20]. However, the accuracy of MEC method depends on the number of nodes. Too many nodes will increase the calculation time and make the calculation more complicated. In addition, the connection relationship of each node in the air-gap needs to be reset at different times due to the change of the relative position between stator and rotor, which

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will further complicate the calculation. The winding function and MEC method are combined in [21], [22] to calculate the armature magnetic field distribution of IPM motors considering the saturation of core, but the method can only obtain the radial flux density of the motors, but cannot calculate the tangential air gap flux density.

In order to balance the calculation accuracy and time of the magnetic field distribution for IPM motors, references [23]–[25] propose an improved subdomain model to predict the air-gap magnetic field of STPM motors considering the saturation of MBs. However, this method is only suitable for STPM motors, and not suitable for other types of IPM motors due to the change of boundary conditions. At the same time, there are few articles that use subdomain method to calculate the air-gap magnetic field for other types of IPM motors considering the saturation of core. This paper proposes a general analytical model which combines subdomain method and MEC method to predict the no-load performance of V-shape IPM motors with any slot-pole combination considering the nonlinearity of MBs. The distinctions between the paper and previous work are as follows.

a) The V-shape permanent magnets (PMs) is equivalent to the form of distribution along the radial and tangential directions to facilitate the establishment of subdomains. This equivalence provides ideas for the analytical derivation of other types of IPM motors with irregular magnets arrangement.
b) The bridge subdomain of V-shape IPM motor is established to make the magnetic field distribution more accurate, in which the permeability of bridges can be obtained by MEC method.
c) The proposed analytical model can be extended to U-shape IPM motors with MBs.

II. The Assumptions and Framework of the Paper

The following assumptions are made in the investigation.

1) The PMs have linear demagnetization characteristics.
2) Ignore the effects of stator and rotor saturation (except for the MBs).
3) The end effect is ignored.

This paper is organized as follows. In Section III, the topology and equivalent process of the V-shape IPM motor are introduced. In Section IV, the analysis method of the equivalent model is presented. Afterwards, the results calculated by the proposed method are then compared with FEA and experiments in Section V. Conclusions are drawn at the end of the paper.

III. Motor Topology and Equivalent Process

The topology of the V-shape IPM motor and its equivalent schematic diagram is shown in Fig. 1. As shown in Fig. 1, the model is established in r-θ coordinate system. It is stipulated that counterclockwise direction is the positive direction of motor rotation.

The main geometric parameters are as follows: R_s, the inner radius of stator core; R_r, the outer radius of rotor core; $h_\mathrm{m}/l_\mathrm{m}$, the thickness/width of PMs; $h_\mathrm{b1}/w_\mathrm{b1}$, the length/width of MBs;

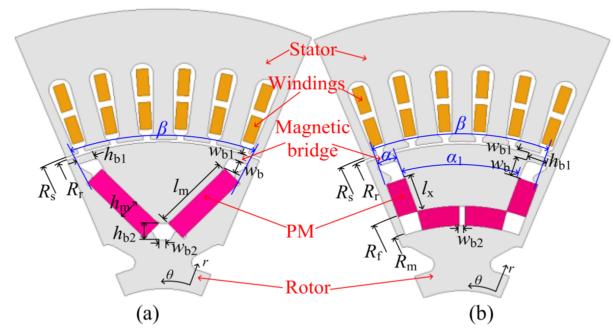


Fig. 1. Topology of V-shape IPM motor and its equivalent schematic diagram. (a) Topology of original V-shape IPM motor. (b) Topology after equivalence.

$h_\mathrm{b2}/w_\mathrm{b2}$, the length/width of air area between PMs; w_b, the width of air area between magnets and MBs; β, the span of PMs; α/α_1, the span of PMs distributed along radial/tangential direction after equivalence; $R_\mathrm{m}/R_\mathrm{f}$, inner/outer radius of PMs distributed along tangential direction after equivalence; l_x, the width of PMs distributed along radial direction after equivalence.

The specific equivalent process is as follows. Firstly, the thickness and span of each magnet should remain unchanged before and after the equivalence. Then, according to the principle that the total flux produced by the magnets does not change, the following equation can be established.

$$2l_\mathrm{m} = 2l_\mathrm{x} + \alpha_1 R_\mathrm{f} - w_\mathrm{b2} \tag{1}$$

$$l_\mathrm{x} = R_\mathrm{r} - R_\mathrm{f} - w_\mathrm{b1} - w_\mathrm{b} \tag{2}$$

$$\alpha_1 = \beta - 2\alpha = \beta - 2[h_\mathrm{m}/(R_\mathrm{f} + l_\mathrm{x})] \tag{3}$$

Finally, combining the above equations, R_f and R_m can be expressed as follows.

$$R_\mathrm{f} = (R_\mathrm{r} - l_\mathrm{m} - w_\mathrm{b1} - w_\mathrm{b} - w_\mathrm{b2}/2)/(1 - \alpha_1/2) \tag{4}$$

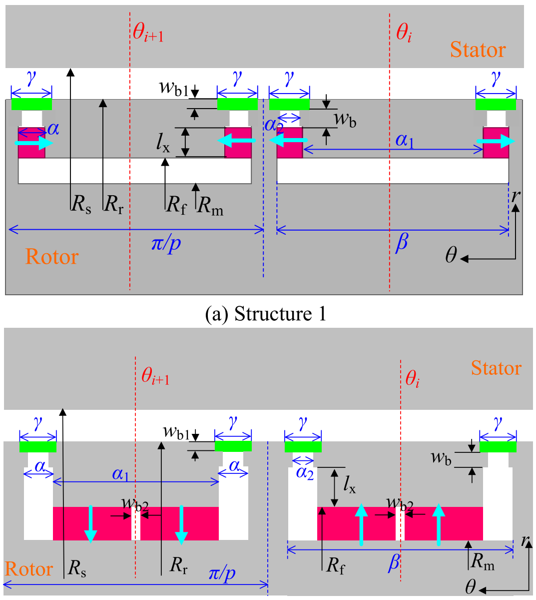
$$R_\mathrm{m} = R_\mathrm{f} - h_\mathrm{m} \tag{5}$$

After the equivalence, the original IPM motor is divided into two corresponding structures, that is, motors with magnets distributed along radial direction (named Structure 1) and tangential direction (named Structure 2), as shown in Fig. 2, where α_2 is the span of air area between magnets and MBs, γ is the span of saturation area in MBs and can be expressed in (7) according to [23].

$$\alpha_2 = h_\mathrm{b1}/(R_\mathrm{r} - w_\mathrm{b1}) \tag{6}$$

$$\gamma = \alpha_2 + 2w_\mathrm{b1}/R_\mathrm{r} \tag{7}$$

It should be noted that the permeability of MBs in the motor with Structure 1 or 2 is the same as that of the original V-shape motor, which can be calculated by MEC method and iterative method. Ignoring the slotting of stator, the MEC of original V-shape motor is shown in Fig. 3, where G_m, G_as1, G_as2, G_b, and G_g are the equivalent permeance of magnet, air area between magnets and MBs, air area between PMs, MBs, and air-gap of the V-shape motor; φ_m, φ_as1, φ_as2, φ_b, and φ_g are the flux of each branch; φ_r is total flux produced by magnets. The equivalent permeance and flux can be calculated according to [26].



(a) Structure 1

(b) Structure 2

Fig. 2. Schematic diagram of the motor after equivalent under a pair of poles.

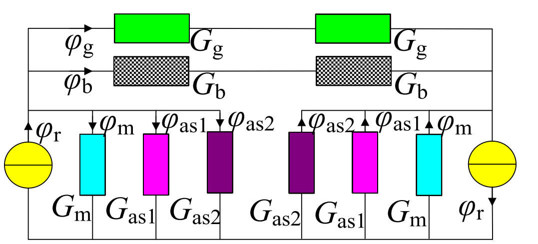


Fig. 3. MEC of the original V-shape IPM motor.

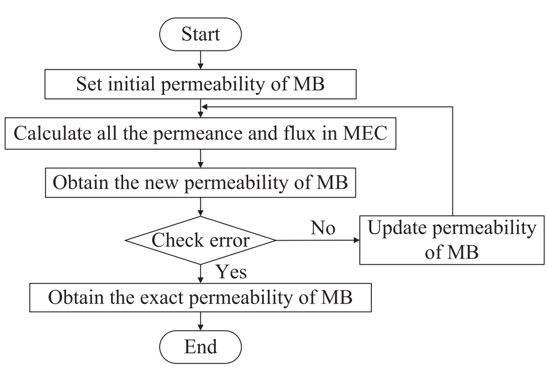


Fig. 4. Flow chart of the iterative method for the calculation of permeability.

The iterative method is employed to obtain the permeability of MBs. Firstly, the initial permeability of bridge is supposed as μ_{b0}, and all the equivalent permeance and flux in MEC can be obtained based on the *B-H* curve of core. Then, the new permeability of MB (μ_{b1}) is calculated by solving the MEC. Check the error $e(\mu_b)$ of μ_{b1} and μ_{b0}, and adjust the initial permeability μ_{b0}. Eventually, repeat the process until the error is satisfied, and obtain the exact permeability of MBs. The calculation process is shown in Fig. 4.

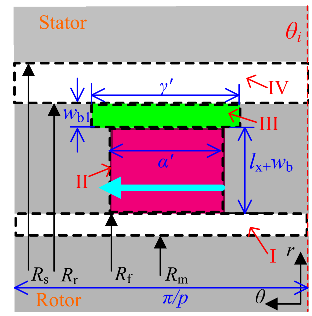


Fig. 5. Simplified model of the motor with Structure 1.

After obtaining the permeability of MBs, the subdomain method is used to obtain the magnetic field distribution of the two structures of motors considering the saturation of MBs. The synthetic magnetic field can be calculated according to superposition principle, and the specific process is as follows.

IV. Analytical Model

A. Establishment of Subdomains for Motor With Structure 1

As can be seen from Fig. 2(a), the magnetization direction of adjacent magnets are the same. In order to facilitate the establishment of subdomains, the following simplifications are made: the magnets with same polarity are combined into a whole magnet with the span of π/p-β+2α by replacing the rotor core between PMs with magnets; and the width of magnets in the radial direction is set to l_x+w_b, ignoring the air area between magnet and MBs. The slotting effect is neglected and the impact of model simplification and slotting on the calculated field distribution will be revised later. The simplified model is shown in Fig. 5, where α' and γ' are the span of magnet and saturation area of MBs in the simplified model respectively, which can be expressed as

$$\alpha' = 2\alpha + \pi/p - \beta \tag{8}$$

$$\gamma' = \alpha' + w_{b1}/R_r \tag{9}$$

It can be seen from Fig. 5 that the motor is divided into 4 subdomains: air slot (I), PM (II), MBs (III) and air-gap (IV). In subdomain I, III and IV, the vector potential *A* is governed by

$$\frac{\partial^2 A}{\partial r^2} + \frac{1}{r}\frac{\partial A}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A}{\partial \theta^2} = 0 \tag{10}$$

and in subdomain II, the vector potential *A* is governed by

$$\frac{\partial^2 A}{\partial r^2} + \frac{1}{r}\frac{\partial A}{\partial r} + \frac{1}{r^2}\frac{\partial^2 A}{\partial \theta^2} = -\mu_0\frac{M}{r} \tag{11}$$

where μ_0 is the permeability of air; *M* is the residual magnetization vector of magnets and can be expressed as

$$M = (-1)^i B_{rem}/\mu_0 \tag{12}$$

where B_{rem} is the remanence of magnets; *i* represents the i^{th} magnet and equal to 1, 2, … 2*p*, as shown in Fig. 2, in which *p* is the number of pole-pairs. Taking the periodicity and symmetry of the model into account, the general solution of vector potential in each subdomain can be expressed as [23]

$$A_I(r,\theta) = \sum_n A_{1\mathrm{n}}\left(\frac{r}{R_\mathrm{f}}\right)^{np}\sin(np\theta) \tag{13}$$

$$\begin{aligned} A_{II}(r,\theta) = {} & A_{02} + B_{02}\ln(r) + B_{rem}r \\ & + \sum_k \left[A_{2\mathrm{k}}\left(\frac{r}{R_\mathrm{f}}\right)^{\frac{k\pi}{\alpha'}} + B_{2\mathrm{k}}\left(\frac{r}{R_\mathrm{r}-w_{\mathrm{b}1}}\right)^{-\frac{k\pi}{\alpha'}}\right] \\ & \cos\left[\frac{k\pi}{\alpha'}\left(\theta - \frac{\pi}{2p} + \frac{\alpha'}{2}\right)\right] \end{aligned} \tag{14}$$

$$\begin{aligned} A_{III}(r,\theta) = {} & A_{03} + B_{03}\ln(r) \\ & + \sum_g \left[A_{3\mathrm{g}}\left(\frac{r}{R_\mathrm{r}-w_{\mathrm{b}1}}\right)^{\frac{g\pi}{\gamma'}} + B_{3\mathrm{g}}\left(\frac{r}{R_\mathrm{r}}\right)^{-\frac{g\pi}{\gamma'}}\right] \\ & \cos\left[\frac{g\pi}{\gamma'}\left(\theta - \frac{\pi}{2p} + \frac{\gamma'}{2}\right)\right] \end{aligned} \tag{15}$$

$$A_{IV}(r,\theta) = \sum_n \left[A_{4\mathrm{n}}\left(\frac{r}{R_\mathrm{r}}\right)^{np} + B_{4\mathrm{n}}\left(\frac{r}{R_\mathrm{s}}\right)^{-np}\right]\sin(np\theta) \tag{16}$$

where n, k, and g are the harmonic orders of different subdomains; $A_{1\mathrm{n}}, A_{02}, B_{02}, A_{2\mathrm{k}}, B_{2\mathrm{k}}, A_{03}, B_{03}, A_{3\mathrm{g}}, B_3, A_{4\mathrm{n}}$, and $B_{4\mathrm{n}}$ are the undetermined coefficients.

In the interface between different subdomains, the vector potential A, radial flux density B_r and tangential magnetic intensity H_θ are continuous. Therefore the boundary conditions between subdomain I and II at R_f are

$$\begin{cases} A_I(R_\mathrm{f},\theta) = A_{II}(R_\mathrm{f},\theta), \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \\ B_{Ir}(R_\mathrm{f},\theta) = B_{IIr}(R_\mathrm{f},\theta), \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \\ H_{I\theta}(R_\mathrm{f},\theta) = H_{II\theta}(R_\mathrm{f},\theta), \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \end{cases} \tag{17}$$

where θ_i is the angular position of i^{th} magnet, and it is specified that it is N pole when i is equal to 1, corresponding to $\theta = 0$. The boundary conditions between subdomain II and III at R_r-$w_{\mathrm{b}1}$ are

$$\begin{cases} A_{II}(R_\mathrm{r}-w_{\mathrm{b}1},\theta) = A_{III}(R_\mathrm{r}-w_{\mathrm{b}1},\theta), \\ \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \\ B_{IIr}(R_\mathrm{r}-w_{\mathrm{b}1},\theta) = B_{IIIr}(R_\mathrm{r}-w_{\mathrm{b}1},\theta), \\ \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \\ H_{II\theta}(R_\mathrm{r}-w_{\mathrm{b}1},\theta) = H_{III\theta}(R_\mathrm{r}-w_{\mathrm{b}1},\theta), \\ \theta_i + \frac{\pi}{2p} - \frac{\alpha'}{2} \le \theta \le \theta_i + \frac{\pi}{2p} + \frac{\alpha'}{2} \end{cases} \tag{18}$$

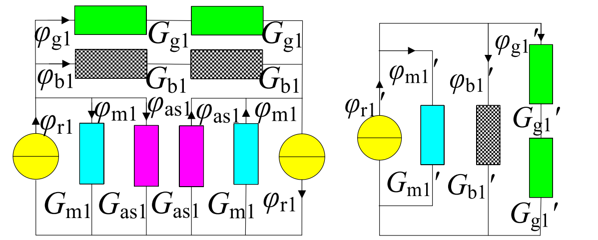


Fig. 6. MEC of the motor with Structure 1. (a) before simplification. (b) after simplification.

Boundary conditions between subdomain III and IV at R_r are

$$\begin{cases} A_{III}(R_\mathrm{r},\theta) = A_{IV}(R_\mathrm{r},\theta), \theta_i + \frac{\pi}{2p} - \frac{\gamma'}{2} \le \theta \le \theta_i \\ \quad + \frac{\pi}{2p} + \frac{\gamma'}{2} \\ B_{IIIr}(R_\mathrm{r},\theta) = B_{IVr}(R_\mathrm{r},\theta), \theta_i + \frac{\pi}{2p} - \frac{\gamma'}{2} \le \theta \le \theta_i \\ \quad + \frac{\pi}{2p} + \frac{\gamma'}{2} \\ H_{III\theta}(R_\mathrm{r},\theta) = H_{IV\theta}(R_\mathrm{r},\theta), \theta_i + \frac{\pi}{2p} - \frac{\gamma'}{2} \le \theta \le \theta_i \\ \quad + \frac{\pi}{2p} + \frac{\gamma'}{2} \end{cases} \tag{19}$$

The boundary condition of subdomain IV at R_s are

$$H_{IV\theta}(R_\mathrm{s},\theta) = 0 \tag{20}$$

According to the Fourier series expansion, the equations with undetermined coefficients can be obtained from the above boundary conditions, which will be given in the appendix. During the equations solving process, the permeability of MBs will change due to the simplification of model, which is different from the original V-shape motor. The permeability of MBs in the simplified motor with Structure 1 can also be obtained by MEC method and iterative method, where the MEC of the motor with Structure 1 is shown in Fig. 6, and the meaning of each parameter is the same as that in Fig. 3.

After obtaining the undetermined coefficients, the analytical formulation of the radial and tangential no-load air-gap magnetic field distribution of the simplified motor with Structure 1 can be expressed as

$$B_{IVr} = \sum_n \left[\frac{npA_{4\mathrm{n}}}{r}\left(\frac{r}{R_\mathrm{r}}\right)^{np} + \frac{npB_{4\mathrm{n}}}{r}\left(\frac{r}{R_\mathrm{s}}\right)^{-np}\right]\cos(np\theta) \tag{21}$$

$$\begin{aligned} B_{IV\theta} = {} & -\sum_n \left[\frac{npA_{4\mathrm{n}}}{R_\mathrm{r}}\left(\frac{r}{R_\mathrm{r}}\right)^{np-1} - \frac{npB_{4\mathrm{n}}}{R_\mathrm{s}}\left(\frac{r}{R_\mathrm{s}}\right)^{-np-1}\right] \\ & \times \sin(np\theta) \end{aligned} \tag{22}$$

In order to consider the influence of model simplification on the magnetic field distribution, the results of (21) and (22) need to be revised. On the one hand, the total flux and leakage flux will change after changing the span and width of PM, and its influence on the amplitude of air-gap flux density can be corrected by (23), where φ_{g1} and $\varphi_{g1}{}'$ can be obtained by solving the MEC shown in Fig. 6. It should be noted that the permeability of bridge in Fig. 6(a) is the same as that in Fig. 3.

$$K_{\text{mod}} = \varphi_{g1}/\varphi'_{g1} \tag{23}$$

On the other hand, considering the existence of non-saturated core between adjacent magnets, the flux path is different due to the permeability of non-saturated core is much greater than that of bridge and air, which will affect the distribution of air-gap magnetic field. Considering the effect of non-saturated core, assume that the radial air gap flux density at the corresponding position of bridge changes linearly, the radial and tangential air-gap magnetic field distribution can be corrected as:

$$B_{IVr\,\text{mod}} = \begin{cases} B_{IVr} \text{ when } |\theta-\theta_i| \le \dfrac{\alpha_1+\alpha-\gamma}{2} \\ (-1)^{i+1}\dfrac{|B_{IVr}|_{\max}}{\gamma}[\theta-(\theta_i-\dfrac{\alpha_1+\alpha+\gamma}{2})] \\ \text{when } \theta_i-\dfrac{\alpha_1+\alpha+\gamma}{2} \le \theta \le \theta_i-\dfrac{\alpha_1+\alpha-\gamma}{2} \\ (-1)^{i}\dfrac{|B_{IVr}|_{\max}}{\gamma}[\theta-(\theta_i+\dfrac{\alpha_1+\alpha+\gamma}{2})] \\ \text{when } \theta_i+\dfrac{\alpha_1+\alpha-\gamma}{2} \le \theta \le \theta_i+\dfrac{\alpha_1+\alpha+\gamma}{2} \\ 0 \quad \text{else where} \end{cases} \tag{24}$$

$$B_{IV\theta\,\text{mod}} = \begin{cases} B_{IV\theta} & \text{when } |\theta-\theta_i| \le \dfrac{\alpha_1+\alpha+\gamma}{2} \\ 0 & \text{elsewhere} \end{cases} \tag{25}$$

Based on the above analysis, the radial and tangential no-load air-gap magnetic field distribution of the original motor with Structure 1 can be expressed as

$$B'_{IVr} = B_{IVr\,\text{mod}} K_{\text{mod}} \tag{26}$$

$$B'_{IV\theta} = B_{IV\theta\,\text{mod}} K_{\text{mod}} \tag{27}$$

B. Establishment of Subdomains for Motor With Structure 2

As can be seen from Fig. 2(b), the model is symmetrical along θ_{i}, so the subdomains can be established in the interval of [θ_{i}, $\theta_{\text{i}}+\pi/2\text{p}$]. In order to facilitate the establishment of subdomains, the following simplifications are made: the MB is ignored and the span of air area is changed to γ, and at the same time, the the span of magnet in tangential direction is extended to $(\alpha+\alpha_1+\gamma)/2$. The simplified model of the motor with Structure 2 is shown in Fig. 7. The stator slotting is neglected and the impact of model simplification and slotting on the calculated field distribution will be revised later.

At this time, the model is divided into 3 subdomains: air-gap (V), air slot (VI), and magnet (VII). The general solution of vector potential in each subdomain can be expressed as [12]

$$A_V(r,\theta) = \sum_n \left[A_{5\text{n}}\left(\frac{r}{R_{\text{s}}}\right)^{np} + B_{5\text{n}}\left(\frac{r}{R_{\text{r}}}\right)^{-np}\right]\sin(np\theta) \tag{28}$$

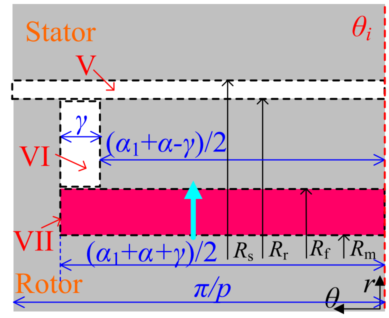


Fig. 7. Simplified model of the motor with Structure 2.

$$A_{VI}(r,\theta) = A_{06} + B_{06}\ln(r) + \sum_m \left[A_{6\text{m}}\left(\frac{r}{R_{\text{r}}}\right)^{\frac{2m\pi}{\gamma}} + B_{6\text{m}}\left(\frac{r}{R_{\text{f}}}\right)^{-\frac{2m\pi}{\gamma}}\right] \times \cos\left[\frac{2m\pi}{\gamma}\left(\theta-\frac{\alpha_1+\alpha}{2}\right)\right] \tag{29}$$

$$A_{VII}(r,\theta) = \sum_k \left[A_{7\text{k}}\left(\frac{r}{R_{\text{f}}}\right)^{\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}} + B_{7\text{k}}\left(\frac{r}{R_{\text{m}}}\right)^{-\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}} - (-1)^k r\frac{4B_{\text{rem}}(\alpha_1+\alpha+\gamma)}{(2k+1)^2\pi^2-(\alpha_1+\alpha+\gamma)^2}\right] \times \sin\left[\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}\theta\right] \tag{30}$$

where *n*, *m*, *k* are the harmonic orders of different subdomains; $A_{5\text{n}}, B_{5\text{n}}, A_{06}, B_{06}, A_{6\text{m}}, B_{6\text{m}}, A_{7\text{k}}$, and $B_{7\text{k}}$ are the undetermined coefficients.

In the interface between different subdomains, the vector potential *A*, radial flux density B_{r} and tangential magnetic intensity H_θ are continuous, respectively. Therefore the boundary conditions between subdomain V and VI at R_{r} are

$$\begin{cases} A_V(R_{\text{r}},\theta) = A_{VI}(R_{\text{r}},\theta), \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i + \dfrac{\alpha+\alpha_1+\gamma}{2} \\ B_{Vr}(R_{\text{r}},\theta) = B_{VIr}(R_{\text{r}},\theta), \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i + \dfrac{\alpha+\alpha_1+\gamma}{2} \\ H_{V\theta}(R_{\text{r}},\theta) = H_{VI\theta}(R_{\text{r}},\theta), \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i + \dfrac{\alpha+\alpha_1+\gamma}{2} \end{cases} \tag{31}$$

Boundary conditions between subdomain VI and VII at R_{f} are

$$\begin{cases} A_{VI}(R_f,\theta) = A_{VII}(R_f,\theta), \\ \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i+\dfrac{\alpha+\alpha_1+\gamma}{2} \\ B_{VIr}(R_f,\theta) = B_{VIIr}(R_f,\theta), \\ \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i+\dfrac{\alpha+\alpha_1+\gamma}{2} \\ H_{VI\theta}(R_f,\theta) = H_{VII\theta}(R_f,\theta), \\ \theta_i+\dfrac{\alpha+\alpha_1-\gamma}{2} \le \theta \le \theta_i+\dfrac{\alpha+\alpha_1+\gamma}{2} \end{cases} \tag{32}$$

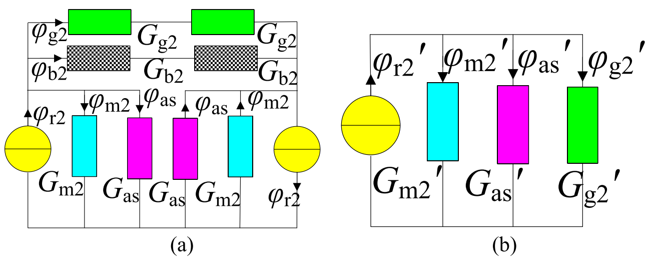


Fig. 8. MEC of the motor with Structure 2. (a) Before simplification. (b) After simplification.

The boundary condition of subdomain V at R_{s} is

$$H_{V\theta}(R_{\mathrm{s}}, \theta) = 0 \tag{33}$$

The boundary condition of subdomain VII at R_{m} is

$$H_{VII\theta}(R_{\mathrm{m}}, \theta) = 0, \theta_i \leq \theta \leq \theta_i + \frac{\alpha_1 + \alpha + \gamma}{2} \tag{34}$$

According to the Fourier series expansion, the equations with undetermined coefficients can be obtained from the above boundary conditions, which will be given in the appendix. After obtaining the coefficients, the analytical formulation of the radial and tangential no-load air-gap magnetic field distribution of the simplified motor with Structure 2 can be expressed as

$$B_{Vr} = \sum_{n} \left[\frac{npA_{5\mathrm{n}}}{r} \left(\frac{r}{R_{\mathrm{s}}} \right)^{np} + \frac{npB_{5\mathrm{n}}}{r} \left(\frac{r}{R_{\mathrm{r}}} \right)^{-np} \right] \cos(np\theta) \tag{35}$$

$$B_{V\theta} = -\sum_{n} \left[\frac{npA_{5\mathrm{n}}}{R_{\mathrm{s}}} \left(\frac{r}{R_{\mathrm{s}}} \right)^{np-1} - \frac{npB_{5\mathrm{n}}}{R_{\mathrm{r}}} \left(\frac{r}{R_{\mathrm{r}}} \right)^{-np-1} \right] \times \sin(np\theta) \tag{36}$$

In order to consider the influence of model simplification on the no-load air-gap magnetic field distribution, the results of (35) and (36) need to be revised. On the one hand, the total flux and leakage flux will change after ignoring the bridge and increasing the span of magnet, and its influence on the amplitude of air-gap flux density can be corrected by (37), where φ_{g2} and $\varphi_{\mathrm{g2}}{}'$ can be obtained by solving the MEC shown in Fig. 8. It should be noted that the permeability of bridge in Fig. 8(a) is the same as that in Fig. 3.

$$K'_{\mathrm{mod}} = \varphi_{g2} / \varphi_{g2}{}' \tag{37}$$

On the other hand, considering the existence of highly saturated bridge in Fig. 2(b), almost all of the flux enters air-gap through the non-saturated rotor core with the interval of [θ_i-(α+α_1-γ)/2, θ_i+(α+α_1-γ)/2]. Since the span of the saturation area of bridge, that is γ, affects the interval of non-saturated core, so the no-load air-gap magnetic field distribution mainly depends on the span of saturation area of bridges. For the simplified model in Fig. 7, the span of air area is set to γ, which is consistent with the span of saturation area of bridges in the original model, so that the flux enters air-gap through the non-saturated rotor core with a span of α+α_1-γ, making the air-gap magnetic field distribution is the same as that in Fig. 2(b).

Based on the above analysis, the radial and tangential no-load air-gap magnetic field distribution of the original motor with Structure 2 can be expressed as

$$B'_{Vr} = B_{Vr} K'_{\mathrm{mod}} \tag{38}$$

$$B'_{V\theta} = B_{V\theta} K'_{\mathrm{mod}} \tag{39}$$

C. Effect of Slotting

The no-load air-gap magnetic field distribution of slotless V-shape IPM motor can be obtained by superposing the field distribution of motors with Structure 1 and 2, which can be expressed as

$$B_r = B'_{IVr} + B'_{Vr} \tag{40}$$

$$B_\theta = B'_{IV\theta} + B'_{V\theta} \tag{41}$$

The conformal mapping method is adopted to consider the slotting effect in this paper. The main idea of the method is to transform the actual slotted air-gap into a slotless air-gap using several conformal transformations, and the complex relative air-gap permeance can be obtained to represent the effect of slotting [27]. Then the radial and tangential components of the air-gap magnetic field of slotted motor can be described as

$$B_{slot\_r} = B_r \lambda_a + B_\theta \lambda_b \tag{42}$$

$$B_{slot\_\theta} = B_\theta \lambda_a - B_r \lambda_b \tag{43}$$

where λ_{a} and λ_{b} are the real and imaginary parts of complex relative air-gap permeance, respectively.

D. No-load Electromagnetic Performance Prediction

The no-load back electromotive force (EMF) of the motor can be obtained according to the distribution of flux linkage

$$E = -\frac{d\psi}{dt} \tag{44}$$

where ψ is the phase flux linkage and can be calculated by

$$\psi = N\Phi = N \int B_{\mathrm{slot}\_r} dS \tag{45}$$

where N is the number of turns per phase in series; Φ is the phase flux, which can be obtained from the radial no-load air-gap magnetic field distribution. The cogging torque of the IPM motor can be expressed as

$$T_{cog} = \frac{L_a r^2}{\mu_0} \int B_{\mathrm{slot}\_r} B_{\mathrm{slot}\_\theta} d\theta \tag{46}$$

where L_{a} is the stack length of the motor.

V. Calculation Results and Verification

To validate the feasibility of the proposed model, the no-load electromagnetic performance of an 8-pole/48-slot V-shape IPM motor are calculated and tested. The configuration of the motor is shown in Fig. 9, and the main parameters are listed in Table I, in which the nonlinear *B-H* curve of the core is shown in Fig. 10.

(a) (b)

Fig. 9. Configuration of the motor. (a) Stator core with windings. (b) Rotor core.

TABLE I
MAIN PARAMETERS OF THE IPM MOTOR

Parameter	Value	Parameter	Value
Rated power	20kW	Rated speed	4500rpm
Number of poles	8	Number of slots(Q)	48
Air-gap length	0.7mm	Magnet remanence	1.05T
Magnet thickness (h_m)	5mm	Rotor outer radius (R_r)	62.8mm
Bridge length (h_{b1})	4mm	Bridge width (w_{b1})	2mm
Bridge length (h_{b2})	5mm	Bridge width (w_{b2})	2mm
Magnet span (β)	0.69	Air slot width (w_b)	4mm
Stack length	75mm	Magnet width (l_m)	20mm
Stator slot opening width	1.8mm	Core material	DW465-50

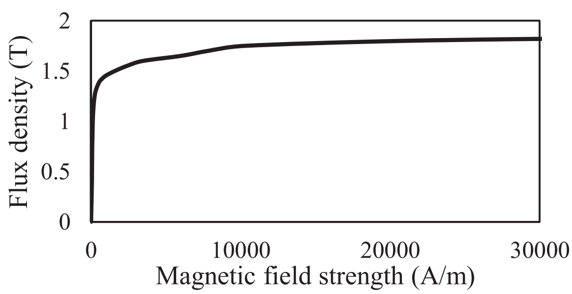


Fig. 10. Nonlinear *B-H* curve of core material.

It should be noted that the prototype not only has MBs between magnets and the outer boundary of rotor core, but also has MBs between the magnet segments with same polarity. Therefore, in order to consider the influence of the MBs between the magnet segments with same polarity in the analytical calculation process, it is necessary to add an additional branch to the MEC shown in Fig. 8.

A. Magnetic Field Distribution in the Air-Gap

The radial and tangential no-load air-gap magnetic field distribution of the prototype obtained by FEA and the proposed analytical method are shown in Fig. 11. In order to quantify the coincidence degree of the no-load air-gap magnetic field distribution calculated by the two methods, the coefficient of determination is used in this paper.

The calculated determination coefficients of radial and tangential air-gap magnetic field distribution are 0.98 and 0.79, respectively. On the one hand, the error is caused by the distortion of air-gap field due to deformations in the circular path when the slotless domain is mapped to the slotted domain in

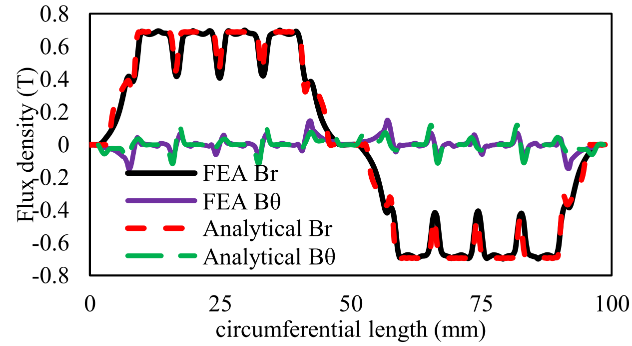
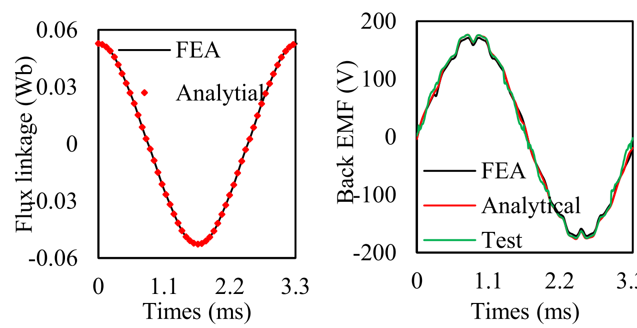


Fig. 11. Radial and tangential no-load air-gap magnetic field distribution.



(a) (b)

Fig. 12. Phase flux linkage and line back EMF of the IPM motor under no-load condition. (a) Phase flux linkage waveform. (b) Line back EMF waveform.

the conformal mapping; on the other hand, the harmonics of the resultant air-gap field obtained by superimposing the field of motors with Structure 1 and 2 will also produce errors compared with the FEA. Another reason is that the saturation degree at the bridge is considered to be uniform in the modeling process, which is different from the actual situation, which will also bring calculation errors.

B. Flux Linkage and No-Load Back EMF

Fig. 12(a) shows the comparison of the calculated A-phase flux linkage obtained by the proposed method and FEA. The calculated determination coefficient of flux linkage is 0.98, indicating that the calculated results are in good agreement.

The no-load line back EMF of the IPM motor at rated speed is tested, and the results are compared with FEA and proposed method, as shown in Fig. 12(b), in which the calculated determination coefficient is 0.96. The harmonic analysis of the back EMF waveforms is shown in Fig. 13, in which that the fundamental amplitudes of the no-load back EMF obtained by FEA, proposed method and experimental test are 168.7V, 172.7V, and 166.9V, respectively. And the corresponding THD are 5.73%, 6.11% and 6.84%, respectively. It can be seen that the proposed method can accurately calculate the no-load back EMF of the motor.

C. Cogging Torque Calculation

The cogging torque of the IPM motor under one tooth pitch calculated by the proposed analytical model is compared with

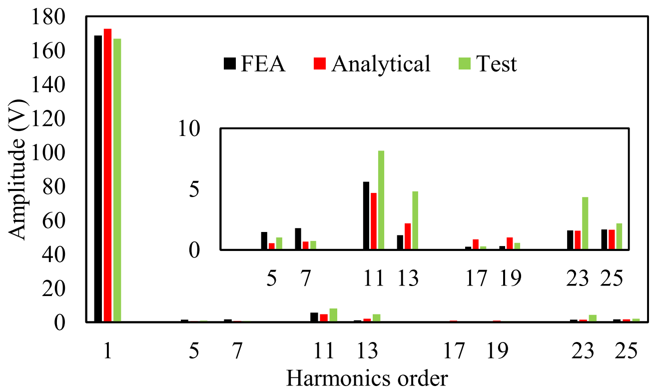


Fig. 13. Harmonic analysis of back EMF waveforms.

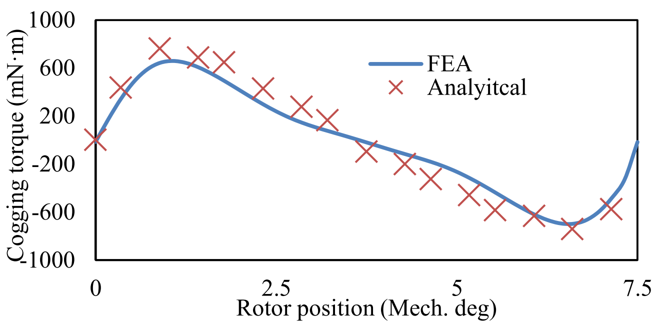


Fig. 14. Cogging torque of the IPM motor.

the FEA, as shown in Fig. 14, in which the calculated determination coefficient is 0.97. It is observed that the cogging torque waveform calculated by the proposed model and FEA has the same variation trend, but the error also exists. The error of cogging torque is caused by the error of no-load air-gap magnetic field distribution. It can be seen from Fig. 14 that the amplitude of cogging torque calculated by FEA is 0.75N·m, while it is 0.7N·m for the proposed model. The calculation error of the amplitude of cogging torque is 6.7%, which is within the allowable range of the engineering. Although there are errors in the calculated cogging torque amplitude, the model can still predict the change trend of cogging torque waveforms of the motors, which shows that the model still has certain practicality for optimizing the motors.

D. Applicable Conditions of the Model

The model shown in Fig. 5 assumes that the flux lines all enter the boundary of MBs along normal direction. However, as the degree of saturation of MBs increases, the direction of the flux line will change, the assumption of $\partial A/\partial\theta = 0$ will produce errors. In order to analyze the effect of MBs saturation on the field distribution of the motor, and obtain the applicable conditions of the model, a sensitivity analysis is performed.

In this paper, the difference in the degree of saturation of MBs is reflected by the difference in relative permeability. The comparison of air-gap flux density calculated by FEA and analytical model under different relative permeability of MBs is shown in Table II, where μ_{rb} and B_{b} represent the relative permeability

TABLE II
Comparison of Air-Gap Flux Density Calculated By FEA and Analytical Model Under Different Relative Permeability of MBs

	B_r	B_θ	B_r	B_θ	B_r	B_θ	B_r	B_θ
μ_{rb}	20		23		26		28	
B_b/T	2.38		2.32		2.26		2.20	
FEA/T	0.333	0.017	0.281	0.013	0.229	0.009	0.176	0.007
Ana/T	0.295	0.014	0.259	0.012	0.214	0.008	0.169	0.007
e/%	11.4	17.6	7.8	7.7	6.5	11.1	4.0	-
μ_{rb}	35		37		39		41	
B_b/T	2.00		1.94		1.88		1.83	
FEA/T	0.125	0.005	0.222	0.009	0.149	0.006	0.096	0.004
Ana/T	0.121	0.005	0.213	0.008	0.145	0.006	0.094	0.004
e/%	3.2	-	4.1	11.1	2.7	-	2.1	-

and flux density of MBs; B_{r} and B_θ represent the amplitude of radial and tangential air-gap flux density; FEA and Ana represent the results calculated by FEA and analytical model, e represents the calculation error.

It can be seen that when μ_{rb} is larger than 28, the calculation results of analytical model are in good agreement with that of FEA, in which the errors are less than 5%; the error will become larger as μ_{rb} continues to decreases. Therefore, for the motor relative permeability of bridge is greater than 28 under no-load conditions, the proposed model can be used to calculate the no-load air-gap magnetic field of the motors; when the relative permeability is less than 28, the calculation error of the model is relatively large.

E. Computational Time

The FEA software used in this paper is Ansys EM19.0. In order to improve the calculation efficiency, 1/8 motor model is calculated, and the number of meshes in the model is 8236. The solution time of FEA is 124 seconds, and that of the proposed method is 18 seconds [Inter Core i5-10400F @ 2.90GHz CPU, 16.0GB RAM].

VI. Conclusion

This paper proposes a novel general analytical model which combines the subdomain method and MEC method to predict the no-load electromagnetic performance of V-shape IPM motors with any slot-pole combination considering the nonlinearity of MBs. The V-shape PMs are equivalent to the form of distribution along the radial and tangential directions to facilitate the establishment of subdomains. In order to consider the saturation of the MB, the MEC method and iterative method are employed to obtain the permeability of MB to establish the boundary conditions. Then, the no-load magnetic field can be predicted by solving the equations based on boundary conditions and the method of conformal mapping is used to consider the effect of slotting.

The results of no-load air-gap magnetic field distribution and line back EMF calculated by the proposed method are in good agreement with FEA and experimental test results. The error of amplitude of no-load air-gap flux density and back EMF between proposed method and FEA/Test is within 3%. The cogging torque calculated by the proposed method has an error in the amplitude compared with FEA, but the method can still predict the variation trend of cogging torque waveforms, which shows that the analytical model still has certain practicality for optimizing the motor. Meanwhile, the applicable conditions of the model is also studied in this paper, the results shows that the proposed model is more suitable for the analysis of the motor whose relative permeability of MBs is greater than 28 under no-load condition.

Compared with FEA, the proposed model can perform fast calculations of no-load electromagnetic performance for V-shape IPM motors, which greatly saves time during the design and optimization stage. In addition, U-shape IPM motors with magnetic bridges can also be analyzed by this model.

APPENDIX

A. Equations of Motor With Structure 1

The following equations can be obtained according to (17)

$$A_{02} + B_{02}\ln(R_{\rm f}) + B_{\rm rem}R_{\rm f} = \frac{1}{\alpha'}\sum_n A_{1\rm n}\Gamma_1 \quad (47)$$

$$A_{2k} + B_{2k}\left(\frac{R_{\rm f}}{R_{\rm r}-w_{\rm b1}}\right)^{-\frac{k\pi}{\alpha'}} = \frac{2}{\alpha'}\sum_n A_{1\rm n}\Gamma_2 \quad (48)$$

$$\frac{A_{1\rm n}n}{\mu_0 R_{\rm f}} = \frac{2}{\pi\mu_{\rm m}}\left(\frac{B_{02}}{R_{\rm f}} + B_{rem}\right)\Gamma_1 + \frac{2}{\pi\mu_{\rm m}} \times \sum_k\left[\frac{A_{2\rm k}k\pi}{\alpha' R_{\rm f}} - \frac{B_{2\rm k}k\pi}{\alpha'(R_{\rm r}-w_{\rm b1})}\left(\frac{R_{\rm f}}{R_{\rm r}-w_{\rm b1}}\right)^{-\frac{k\pi}{\alpha'}-1}\right]\Gamma_2 \quad (49)$$

The following equations can be obtained according to (18)

$$A_{03}+B_{03}\ln(R_{\rm r}-w_{\rm b1}) = \frac{1}{\gamma'}\sum_k\left[A_{2\rm k}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm f}}\right)^{\frac{k\pi}{\alpha'}} + B_{2\rm k}\right]\Gamma_3 + \frac{\alpha'}{\gamma'}[A_{02} + B_{02}\ln(R_{\rm r}-w_{\rm b1}) + B_{rem}(R_{\rm r}-w_{\rm b1})] \quad (50)$$

$$A_{3\rm g}+B_{3\rm g}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm r}}\right)^{-\frac{g\pi}{\gamma'}} = \frac{2}{\gamma'}\sum_k\left[A_{2\rm k}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm f}}\right)^{\frac{k\pi}{\alpha'}} + B_{2\rm k}\right]\Gamma_5 + \frac{2}{\gamma'}[A_{02} + B_{02}\ln(R_{\rm r}-w_{\rm b1}) + B_{rem}(R_{\rm r}-w_{\rm b1})]\Gamma_4 \quad (51)$$

$$\frac{1}{\mu_{\rm m}}\left(\frac{B_{02}}{R_{\rm r}-w_{\rm b1}} + B_{rem}\right) = \frac{B_{03}}{\mu_{\rm b}(R_{\rm r}-w_{\rm b1})} + \frac{1}{\alpha'\mu_{\rm b}}\sum_g\left[\frac{A_{3\rm g}g\pi}{\gamma'(R_{\rm r}-w_{\rm b1})} - \frac{B_{3\rm g}g\pi}{\gamma' R_{\rm r}}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm r}}\right)^{-\frac{g\pi}{\gamma'}-1}\right]\Gamma_4 \quad (52)$$

$$\frac{1}{\mu_{\rm m}}\left[\frac{A_{2\rm k}k\pi}{\alpha' R_{\rm f}}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm f}}\right)^{\frac{k\pi}{\alpha'}-1} - \frac{B_{2\rm k}k\pi}{\alpha'(R_{\rm r}-w_{\rm b1})}\right] = \frac{2B_{03}\Gamma_3}{\mu_{\rm b}\alpha'(R_{\rm r}-w_{\rm b1})} + \frac{2}{\alpha'\mu_{\rm b}} \times \sum_g\left[\frac{A_{3\rm g}g\pi}{\gamma'}\frac{1}{R_{\rm r}-w_{\rm b1}} - \frac{B_{3\rm g}g\pi}{\gamma' R_{\rm r}}\left(\frac{R_{\rm r}-w_{\rm b1}}{R_{\rm r}}\right)^{-\frac{g\pi}{\gamma'}-1}\right]\Gamma_5 \quad (53)$$

The following equations can be obtained according to (19)

$$A_{03} + B_{03}\ln(R_{\rm r}) = \frac{1}{\gamma'}\sum_n\left[A_{4\rm n} + B_{4\rm n}\left(\frac{R_{\rm r}}{R_{\rm s}}\right)^{-np}\right]\Gamma_6 \quad (54)$$

$$A_{3\rm g}\left(\frac{R_{\rm r}}{R_{\rm r}-w_{\rm b1}}\right)^{\frac{g\pi}{\gamma'}} + B_{3\rm g} = \frac{2}{\gamma'}\sum_n\left[A_{4\rm n} + B_{4\rm n}\left(\frac{R_{\rm r}}{R_{\rm s}}\right)^{-np}\right]\Gamma_7 \quad (55)$$

$$-\frac{1}{\mu_0}\left[\frac{nA_{4n}}{R_{\rm r}} - \frac{nB_{4n}}{R_{\rm s}}\left(\frac{R_{\rm r}}{R_{\rm s}}\right)^{-np-1}\right] = -\frac{2}{\mu_{\rm b}\pi}\frac{B_{03}}{R_{\rm r}}\Gamma_6 - \frac{2}{\mu_{\rm b}\pi}\sum_g\left[\frac{A_{3\rm g}g\pi}{\gamma'(R_{\rm r}-w_{\rm b1})}\left(\frac{R_{\rm r}}{R_{\rm r}-w_{\rm b1}}\right)^{\frac{g\pi}{\gamma'}-1} - \frac{B_{3\rm g}g\pi}{\gamma' R_{\rm r}}\right]\Gamma_7 \quad (56)$$

The following equation can be obtained according to (20)

$$\frac{A_{4n}}{R_r}\left(\frac{R_s}{R_r}\right)^{np-1} = \frac{B_{4n}}{R_s} \quad (57)$$

where $\mu_{\rm b}$ and $\mu_{\rm m}$ are the permeability of bridge and magnet, and

$$\Gamma_1 = \int_{\frac{\pi}{2p}-\frac{\alpha'}{2}}^{\frac{\pi}{2p}+\frac{\alpha'}{2}} \sin(np\theta)d\theta \quad (58)$$

$$\Gamma_2 = \int_{\frac{\pi}{2p}-\frac{\alpha'}{2}}^{\frac{\pi}{2p}+\frac{\alpha'}{2}} \cos\left[\frac{k\pi}{\alpha'}\left(\theta - \frac{\pi}{2p} + \frac{\alpha'}{2}\right)\right]\sin(np\theta)d\theta \quad (59)$$

$$\Gamma_3 = \int_{\frac{\pi}{2p}-\frac{\alpha'}{2}}^{\frac{\pi}{2p}+\frac{\alpha'}{2}} \cos\left[\frac{k\pi}{\alpha'}\left(\theta - \frac{\pi}{2p} + \frac{\alpha'}{2}\right)\right]d\theta \quad (60)$$

$$\Gamma_4 = \int_{\frac{\pi}{2p}-\frac{\alpha'}{2}}^{\frac{\pi}{2p}+\frac{\alpha'}{2}} \cos\left[\frac{g\pi}{\gamma'}\left(\theta - \frac{\pi}{2p} + \frac{\gamma'}{2}\right)\right]d\theta \quad (61)$$

$$\Gamma_5 = \int_{\frac{\pi}{2p}-\frac{\alpha'}{2}}^{\frac{\pi}{2p}+\frac{\alpha'}{2}} \cos\left[\frac{k\pi}{\alpha'}\left(\theta - \frac{\pi}{2p} + \frac{\alpha'}{2}\right)\right] \times \cos\left[\frac{g\pi}{\gamma'}\left(\theta - \frac{\pi}{2p} + \frac{\gamma'}{2}\right)\right]d\theta \quad (62)$$

$$\Gamma_6 = \int_{\frac{\pi}{2p}-\frac{\gamma'}{2}}^{\frac{\pi}{2p}+\frac{\gamma'}{2}} \sin(np\theta)d\theta \quad (63)$$

$$\Gamma_7 = \int_{\frac{\pi}{2p}-\frac{\gamma'}{2}}^{\frac{\pi}{2p}+\frac{\gamma'}{2}} \sin(np\theta)\cos\left[\frac{g\pi}{\gamma'}\left(\theta - \frac{\pi}{2p} + \frac{\gamma'}{2}\right)\right]d\theta \quad (64)$$

B. Equations of Motor With Structure 2

The following equations can be obtained according to (31)

$$A_{06} + B_{06} \ln(R_{\mathrm{r}}) = \sum_n \frac{1}{\gamma} \left[A_{5\mathrm{n}} \left(\frac{R_{\mathrm{r}}}{R_{\mathrm{s}}} \right)^{np} + B_{5\mathrm{n}} \right] \Gamma_8 \quad (65)$$

$$A_{6\mathrm{m}} + B_{6\mathrm{m}} \left(\frac{R_{\mathrm{f}}}{R_{\mathrm{r}}} \right)^{\frac{2m\pi}{\alpha}} = \sum_n \frac{2}{\gamma} \left[A_{5\mathrm{n}} \left(\frac{R_{\mathrm{r}}}{R_{\mathrm{s}}} \right)^{np} + B_{5\mathrm{n}} \right] \Gamma_9 \quad (66)$$

$$-\frac{n}{\mu_0} \left[A_{5\mathrm{n}} \left(\frac{R_{\mathrm{r}}}{R_{\mathrm{s}}} \right)^{np} - B_{5\mathrm{n}} \right] = \frac{2}{\pi \mu_m} \left[(-1)^n - 1 \right] \times \left[B_{06}\Gamma_8 + \sum_m \frac{2\pi m}{\gamma} \left[A_{6\mathrm{m}} - B_{6\mathrm{m}} \left(\frac{R_{\mathrm{f}}}{R_{\mathrm{r}}} \right)^{\frac{2m\pi}{\gamma}} \right] \Gamma_9 \right] \quad (67)$$

The following equations can be obtained according to (32)

$$A_{06} + B_{06} \ln(R_{\mathrm{f}}) = \sum_k \frac{1}{\gamma} \left[A_{7\mathrm{k}} + B_{7\mathrm{k}} \left(\frac{R_{\mathrm{m}}}{R_{\mathrm{f}}} \right)^{\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}} - R_{\mathrm{f}}\Delta_k \right] \Gamma_{10} \quad (68)$$

$$A_{6\mathrm{m}} \left(\frac{R_{\mathrm{f}}}{R_{\mathrm{r}}} \right)^{\frac{2m\pi}{\gamma}} + B_{6\mathrm{m}} = \sum_k \frac{2}{\gamma} \left[A_{7\mathrm{k}} + B_{7\mathrm{k}} \left(\frac{R_{\mathrm{m}}}{R_{\mathrm{f}}} \right)^{\frac{\pi(2k+1)}{\alpha_1+\alpha+\gamma}} - R_{\mathrm{f}}\Delta_k \right] \Gamma_{11} \quad (69)$$

$$\frac{1}{\mu_{\mathrm{m}}} \sum_k \left[\frac{A_{7\mathrm{k}}(2k+1)\pi}{\alpha_1+\alpha+\gamma} - \frac{B_{7\mathrm{k}}(2k+1)\pi}{\alpha_1+\alpha+\gamma} \times \left(\frac{R_{\mathrm{m}}}{R_{\mathrm{f}}} \right)^{\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}} - R_{\mathrm{f}}\Delta_k \right] = \frac{4}{\mu_0(\alpha_1+\alpha+\gamma)} \cdot \left[B_{06}\Gamma_{10} + \sum_m \frac{2m\pi}{\gamma} \times \left[A_{6\mathrm{m}} \left(\frac{R_{\mathrm{f}}}{R_{\mathrm{r}}} \right)^{\frac{2m\pi}{\gamma}} - B_{6\mathrm{m}} \right] \Gamma_{11} \right] \quad (70)$$

The following equation can be obtained according to (33)

$$\frac{A_{5\mathrm{n}}}{R_{\mathrm{s}}} = \frac{B_{5\mathrm{n}}}{R_{\mathrm{r}}} \left(\frac{R_{\mathrm{s}}}{R_{\mathrm{r}}} \right)^{-np-1} \quad (71)$$

The following equation can be obtained according to (34)

$$\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma} \left[\frac{A_{7\mathrm{k}}}{R_{\mathrm{f}}} \left(\frac{R_{\mathrm{m}}}{R_{\mathrm{f}}} \right)^{\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma}-1} - \frac{B_{7\mathrm{k}}}{R_{\mathrm{m}}} \right] = \Delta_k \quad (72)$$

where

$$\Delta_k = (-1)^k \frac{4B_{\mathrm{rem}}(\alpha_1+\alpha+\gamma)}{(2k+1)^2\pi^2 - (\alpha_1+\alpha+\gamma)^2} \quad (73)$$

$$\Gamma_8 = \int_{(\alpha_1+\alpha-\gamma)/2}^{(\alpha_1+\alpha+\gamma)/2} \sin(np\theta) d\theta \quad (74)$$

$$\Gamma_9 = \int_{(\alpha_1+\alpha-\gamma)/2}^{(\alpha_1+\alpha+\gamma)/2} \sin(np\theta) \cos \left[\frac{2m\pi}{\gamma} \left(\theta - \frac{\alpha_1+\alpha}{2} \right) \right] d\theta \quad (75)$$

$$\Gamma_{10} = \int_{(\alpha_1+\alpha-\gamma)/2}^{(\alpha_1+\alpha+\gamma)/2} \sin \left[\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma} \theta \right] d\theta \quad (76)$$

$$\Gamma_{11} = \int_{(\alpha_1+\alpha-\gamma)/2}^{(\alpha_1+\alpha+\gamma)/2} \sin \left[\frac{(2k+1)\pi}{\alpha_1+\alpha+\gamma} \theta \right] \times \cos \left[\frac{2m\pi}{\gamma} \left(\theta - \frac{\alpha_1+\alpha}{2} \right) \right] d\theta \quad (77)$$

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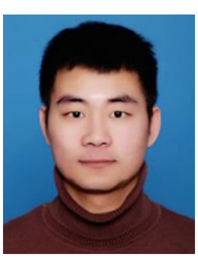

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